

## ASCII Table for Competitive Programming

| Dec | Hex  | Char | Description         |
|-----|------|------|---------------------|
| 32  | 0x20 |      | Space               |
| 33  | 0x21 | !    | Exclamation mark !  |
| 34  | 0x22 | "    | Double quote "      |
| 35  | 0x23 | #    | Hash #              |
| 36  | 0x24 | \$   | Dollar \$           |
| 37  | 0x25 | %    | Percent %           |
| 38  | 0x26 | &    | Ampersand &         |
| 39  | 0x27 | '    | Single quote '      |
| 40  | 0x28 | (    | Left parenthesis (  |
| 41  | 0x29 | )    | Right parenthesis ) |
| 42  | 0x2A | *    | Asterisk *          |
| 43  | 0x2B | +    | Plus +              |
| 44  | 0x2C | ,    | Comma ,             |
| 45  | 0x2D | -    | Hyphen -            |
| 46  | 0x2E | .    | Period .            |
| 47  | 0x2F | /    | Slash /             |
| 48  | 0x30 | 0    | Zero                |
| 49  | 0x31 | 1    |                     |
| 50  | 0x32 | 2    |                     |
| 51  | 0x33 | 3    |                     |
| 52  | 0x34 | 4    |                     |
| 53  | 0x35 | 5    |                     |
| 54  | 0x36 | 6    |                     |
| 55  | 0x37 | 7    |                     |
| 56  | 0x38 | 8    |                     |
| 57  | 0x39 | 9    | Nine                |
| 58  | 0x3A | :    | Colon :             |
| 59  | 0x3B | ;    | Semicolon ;         |
| 60  | 0x3C | <    | Less than <         |
| 61  | 0x3D | =    | Equals =            |
| 62  | 0x3E | >    | Greater than >      |
| 63  | 0x3F | ?    | Question mark ?     |
| 64  | 0x40 | @    | At symbol @         |
| 65  | 0x41 | A    | A                   |
| 66  | 0x42 | B    |                     |
| 67  | 0x43 | C    |                     |
| 68  | 0x44 | D    |                     |
| 69  | 0x45 | E    |                     |
| 70  | 0x46 | F    |                     |
| 71  | 0x47 | G    |                     |

|     |      |   |                 |
|-----|------|---|-----------------|
| 72  | 0x48 | H |                 |
| 73  | 0x49 | I |                 |
| 74  | 0x4A | J |                 |
| 75  | 0x4B | K |                 |
| 76  | 0x4C | L |                 |
| 77  | 0x4D | M |                 |
| 78  | 0x4E | N |                 |
| 79  | 0x4F | O |                 |
| 80  | 0x50 | P |                 |
| 81  | 0x51 | Q |                 |
| 82  | 0x52 | R |                 |
| 83  | 0x53 | S |                 |
| 84  | 0x54 | T |                 |
| 85  | 0x55 | U |                 |
| 86  | 0x56 | V |                 |
| 87  | 0x57 | W |                 |
| 88  | 0x58 | X |                 |
| 89  | 0x59 | Y |                 |
| 90  | 0x5A | Z | Z               |
| 91  | 0x5B | [ | Left bracket [  |
| 92  | 0x5C | \ | Backslash \     |
| 93  | 0x5D | ] | Right bracket ] |
| 94  | 0x5E | ^ | Caret ^         |
| 95  | 0x5F | _ | Underscore _    |
| 96  | 0x60 | ` | Backtick `      |
| 97  | 0x61 | a | a               |
| 98  | 0x62 | b |                 |
| 99  | 0x63 | c |                 |
| 100 | 0x64 | d |                 |
| 101 | 0x65 | e |                 |
| 102 | 0x66 | f |                 |
| 103 | 0x67 | g |                 |
| 104 | 0x68 | h |                 |
| 105 | 0x69 | i |                 |
| 106 | 0x6A | j |                 |
| 107 | 0x6B | k |                 |
| 108 | 0x6C | l |                 |
| 109 | 0x6D | m |                 |
| 110 | 0x6E | n |                 |
| 111 | 0x6F | o |                 |
| 112 | 0x70 | p |                 |
| 113 | 0x71 | q |                 |
| 114 | 0x72 | r |                 |
| 115 | 0x73 | s |                 |

|     |      |   |               |
|-----|------|---|---------------|
| 116 | 0x74 | t |               |
| 117 | 0x75 | u |               |
| 118 | 0x76 | v |               |
| 119 | 0x77 | w |               |
| 120 | 0x78 | x |               |
| 121 | 0x79 | y |               |
| 122 | 0x7A | z | z             |
| 123 | 0x7B | { | Left brace {  |
| 124 | 0x7C |   | Pipe          |
| 125 | 0x7D | } | Right brace } |
| 126 | 0x7E | ~ | Tilde ~       |

### Common ASCII Values & Tricks for CP:

'0'-'9' -> 48-57 | 'A'-'Z' -> 65-90 | 'a'-'z' -> 97-122

'A' - 'a' = -32 (convert lowercase to uppercase)

'a' - 'A' = 32 (convert uppercase to lowercase)

'0' char to int -> subtract 48

Space = 32, Newline = 10 ('\n'), Tab = 9 ('\t')